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Inventor(s)	Assouline; Avihay et al.

Method, system, and medium for augmented reality product recommendations

Abstract

Aspects of the present disclosure involve a system comprising a computer-readable storage medium storing a program and a method for performing operations comprising: receiving a video that includes a depiction of a real-world object in a real-world environment; determining a classification for the real-world environment by processing the real-world object depicted in the video; selecting an augmented reality (AR) item based on the classification of the real-world environment and the real-world object depicted in the video; modifying pixels corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object; and adding the AR item to the modified video at a display position corresponding to the modified pixels.

Inventors: **Assouline; Avihay (Tel Aviv, IL), Berger; Itamar (Hod Hasharon, IL), Dudovitch; Gal (Tel Aviv, IL), Harel; Peleg (Ramat Gan, IL), Sasson; Gal (Kibbutz Ayyelet Hashahar, IL)**

Applicant: **Snap Inc.** (Santa Monica, CA)

Family ID: **85157278**

Assignee: **Snap Inc. (Santa Monica, CA)**

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Primary Examiner: Thein; Marissa

Assistant Examiner: Mitros; Anna Mae

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates generally to providing augmented reality experiences.
BACKGROUND
(2) Augmented-Reality (AR) is a modification of a virtual environment. For example, in Virtual Reality (VR), a user is completely immersed in a virtual world, whereas in AR, the user is immersed in a world where virtual objects are combined or superimposed on the real world. An AR system aims to generate and present virtual objects that interact realistically with a real-world environment and with each other. Examples of AR applications can include single or multiple player video games, instant messaging systems, and the like.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

(2) FIG. **1** is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

(3) FIG. **2** is a block diagram illustrating further details regarding the messaging system, according to some examples.

(4) FIG. **3** is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

(5) FIG. **4** is a diagrammatic representation of a message, in accordance with some examples.

(6) FIG. **5** is a block diagram showing an example AR recommendation system, according to example examples.

(7) FIGS. **6-8** are diagrammatic representations of outputs of the AR recommendation system, in accordance with some examples.

(8) FIG. **9** is a flowchart illustrating example operations of the AR recommendation system, according to examples.

(9) FIG. **10** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

(10) FIG. **11** is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

(11) The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples. It will be evident, however, to those skilled in the art, that examples may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

(12) Typically, virtual reality (VR) and augmented reality (AR) systems allow users to add augmented reality elements to their environment (e.g., captured image data corresponding to a user's surroundings). Such systems can recommend AR elements based on various external factors, such as a current geographical location of the user and various other contextual clues. Some AR systems allow a user to capture a video of a room and select from a list of available AR elements to add to a room to see how the selected AR element looks in the room. These systems allow a user to preview how a physical item looks at a particular location in a user's environment, which simplifies the purchasing process. However, these systems require a user to manually select which AR elements to display within the captured video. Specifically, the user of these systems has to spend a great deal of effort searching through and navigating multiple user interfaces and pages of information to identify an item of interest. Then the user has to manually position the selected item within view. These tasks can be daunting and time consuming, which detracts from the overall interest in using these systems and results in wasted resources.

(13) Sometimes the AR element added by the user can interfere with real-world objects depicted in the video. For example, the user can desire to place an AR coffee table in a video that already includes a coffee table. However, the user is limited to placing the AR coffee table in an area that has free space; otherwise, the AR coffee table will overlap the real-world objects (e.g., the real-world coffee table). Overlapping the AR coffee table with the real-world objects breaks the illusion

of having the AR coffee table appear as though it is part of the real-world environment. Also, it may be difficult to fully appreciate how the AR coffee table would look in place of the real-world coffee table if the AR coffee table is placed on top of the real-world coffee table. This severely limits the functionality of the typical AR systems and detracts from the overall interest in using these systems.

(14) The disclosed techniques improve the efficiency of using an electronic device that implements or otherwise accesses an AR/VR system by intelligently automatically determining what room or real-world environment is within view of a camera and automatically recommending AR elements to display within the camera view, such as for a user to purchase corresponding physical or electronically consumable items (e.g., video items, music items, or video game items). The disclosed techniques can remove real-world objects depicted in the video to make room for the AR elements. Specifically, the disclosed techniques receive a video that includes a depiction of a real-world object in a real-world environment. The disclosed techniques determine a classification for the real-world environment by processing the real-world object depicted in the video and selecting an AR item based on the classification of the real-world environment and the real-world object depicted in the video. The disclosed techniques modify pixels corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object. The disclosed techniques add the AR item to the modified video at a display position corresponding to the modified pixels. This improves the overall user experience and enhances the illusion of the AR elements being part of the real-world environment depicted in the video.

(15) In some cases, the disclosed techniques train a neural network classifier to determine the real-world classification. To train the neural network classifier, the disclosed techniques receive training data comprising a plurality of training images and ground truth room classifications for each of the plurality of training images, each of the plurality of training images depicting a different real-world environment (e.g., a room in a home), and apply the neural network classifier to a first training image of the plurality of training images to estimate a classification of the real-world environment depicted in the first training image. The disclosed techniques compute a deviation between the estimated classification and the ground truth classification associated with the first training image and update parameters of the neural network classifier based on the computed deviation.

(16) In this way, the disclosed techniques can select and automatically display one or more AR elements corresponding to items available for purchase in the current image or video without further input from a user. This improves the overall experience of the user in using the electronic device and reduces the overall amount of system resources needed to accomplish a task.

(17) Networked Computing Environment

(18) FIG. **1** is a diagrammatic representation of a networked environment of a messaging system **100** in which the present disclosure may be deployed, in accordance with some examples. The messaging system **100** includes multiple instances of a client device **102**, each of which hosts a number of applications, including a messaging client **104** and other external applications **109** (e.g., third-party applications). Each messaging client **104** is communicatively coupled to other instances of the messaging client **104** (e.g., hosted on respective other client devices **102**), a messaging server system **108** and external app(s) servers **110** via a network **112** (e.g., the Internet). A messaging client **104** can also communicate with locally-hosted third-party applications (also referred to as “external applications” and “external apps”) **109** using Applications Program Interfaces (APIs).

(19) The client device **102** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the client device **102** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The client device **102** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device

(e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the disclosed operations. Further, while only a single client device **102** is illustrated, the term "client device" shall also be taken to include a collection of machines that individually or jointly execute the disclosed operations.

(20) In one example, the client device **102** can include AR glasses or an AR headset in which virtual content is displayed within lenses of the glasses while a user views a real-world environment through the lenses. For example, an image can be presented on a transparent display that allows a user to simultaneously view content presented on the display and real-world objects. In some cases, the AR headset can remove real-world objects viewed through the lenses of the AR headset according to the disclosed techniques, such as by blurring a region of the lenses where the real-world object is being viewed and overlaying an AR object.

(21) A messaging client **104** is able to communicate and exchange data with other messaging clients **104** and with the messaging server system **108** via the network **112**. The data exchanged between messaging clients **104**, and between a messaging client **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

(22) The messaging server system **108** provides server-side functionality via the network **112** to a particular messaging client **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client **104** or by the messaging server system **108**, the location of certain functionality either within the messaging client **104** or the messaging server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108** but to later migrate this technology and functionality to the messaging client **104** where a client device **102** has sufficient processing capacity.

(23) The messaging server system **108** supports various services and operations that are provided to the messaging client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client **104**.

(24) Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **116** is coupled to, and provides a programmatic interface to, application servers **114**. The application servers **114** are communicatively coupled to a database server **120**, which facilitates access to a database **126** that stores data associated with messages processed by the application servers **114**. Similarly, a web server **128** is coupled to the application servers **114**, and provides web-based interfaces to the application servers **114**. To this end, the web server **128** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(25) The Application Program Interface (API) server **116** receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application servers **114**. Specifically, the API server **116** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client **104** in order to invoke functionality of the application servers **114**. The API server **116** exposes various functions supported by the application servers **114**, including account registration, login functionality, the sending of messages, via the application servers **114**, from a particular messaging client **104** to another messaging client **104**, the sending of media files (e.g., images or video) from a messaging client **104** to a messaging server **118**, and for possible access by another messaging client **104**, the settings of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such

collections, the retrieval of messages and content, the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph), the location of friends within a social graph, and opening an application event (e.g., relating to the messaging client **104**).

(26) The application servers **114** host a number of server applications and subsystems, including for example a messaging server **118**, an image processing server **122**, and a social network server **124**. The messaging server **118** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client **104**. Other processor- and memory-intensive processing of data may also be performed server-side by the messaging server **118**, in view of the hardware requirements for such processing.

(27) The application servers **114** also include an image processing server **122** that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server **118**.

(28) Image processing server **122** is used to implement scan functionality of the augmentation system **208** (shown in FIG. **2**). Scan functionality includes activating and providing one or more augmented reality experiences on a client device **102** when an image is captured by the client device **102**. Specifically, the messaging client **104** on the client device **102** can be used to activate a camera. The camera displays one or more real-time images or a video to a user along with one or more icons or identifiers of one or more augmented reality experiences. The user can select a given one of the identifiers to launch the corresponding augmented reality experience or perform a selected image modification (e.g., launching an AR shopping experience, as discussed in connection with FIGS. **5-9** below).

(29) The social network server **124** supports various social networking functions and services and makes these functions and services available to the messaging server **118**. To this end, the social network server **124** maintains and accesses an entity graph **308** (as shown in FIG. **3**) within the database **126**. Examples of functions and services supported by the social network server **124** include the identification of other users of the messaging system **100** with which a particular user has relationships or is "following," and also the identification of other entities and interests of a particular user.

(30) Returning to the messaging client **104**, features and functions of an external resource (e.g., a third-party application **109** or applet) are made available to a user via an interface of the messaging client **104**. The messaging client **104** receives a user selection of an option to launch or access features of an external resource (e.g., a third-party resource), such as external apps **109**. The external resource may be a third-party application (external apps **109**) installed on the client device **102** (e.g., a "native app"), or a small-scale version of the third-party application (e.g., an "applet") that is hosted on the client device **102** or remote of the client device **102** (e.g., on external resource or app(s) servers **110**). The small-scale version of the third-party application includes a subset of features and functions of the third-party application (e.g., the full-scale, native version of the third-party standalone application) and is implemented using a markup-language document. In one example, the small-scale version of the third-party application (e.g., an "applet") is a web-based, markup-language version of the third-party application and is embedded in the messaging client **104**. In addition to using markup-language documents (e.g., a .*ml file), an applet may incorporate a scripting language (e.g., a .*js file or a .json file) and a style sheet (e.g., a .*ss file).

(31) In response to receiving a user selection of the option to launch or access features of the external resource (e.g., external app **109**), the messaging client **104** determines whether the selected external resource is a web-based external resource or a locally-installed external application. In some cases, external applications **109** that are locally installed on the client device **102** can be

launched independently of and separately from the messaging client **104**, such as by selecting an icon, corresponding to the external application **109**, on a home screen of the client device **102**. Small-scale versions of such external applications can be launched or accessed via the messaging client **104** and, in some examples, no or limited portions of the small-scale external application can be accessed outside of the messaging client **104**. The small-scale external application can be launched by the messaging client **104** receiving, from an external app(s) server **110**, a markup-language document associated with the small-scale external application and processing such a document.

(32) In response to determining that the external resource is a locally-installed external application **109**, the messaging client **104** instructs the client device **102** to launch the external application **109** by executing locally-stored code corresponding to the external application **109**. In response to determining that the external resource is a web-based resource, the messaging client **104** communicates with the external app(s) servers **110** to obtain a markup-language document corresponding to the selected resource. The messaging client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client **104**.

(33) The messaging client **104** can notify a user of the client device **102**, or other users related to such a user (e.g., "friends"), of activity taking place in one or more external resources. For example, the messaging client **104** can provide participants in a conversation (e.g., a chat session) in the messaging client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using a respective messaging client messaging clients **104**, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

(34) The messaging client **104** can present a list of the available external resources (e.g., third-party or external applications **109** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the external applications **109** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

(35) System Architecture

(36) FIG. **2** is a block diagram illustrating further details regarding the messaging system **100**, according to some examples. Specifically, the messaging system **100** is shown to comprise the messaging client **104** and the application servers **114**. The messaging system **100** embodies a number of subsystems, which are supported on the client-side by the messaging client **104** and on the sever-side by the application servers **114**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **208**, a map system **210**, a game system **212**, and an external resource system **220**.

(37) The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the messaging client **104** and the messaging server **118**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

(38) The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client **104**.
(39) The collection management system **204** furthermore includes a curation interface **206** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **206** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.
(40) The augmentation system **208** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system **208** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The augmentation system **208** operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client **104** based on a geolocation of the client device **102**. In another example, the augmentation system **208** operatively supplies a media overlay to the messaging client **104** based on other information, such as social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay may include text, a graphical element, or an image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **208** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **126** and accessed through the database server **120**.
(41) In some examples, the augmentation system **208** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **208** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.
(42) In other examples, the augmentation system **208** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **208** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time. The augmentation system **208** communicates with the image processing server **122** to obtain augmented reality experiences and presents identifiers of such experiences in one or more user interfaces (e.g., as icons over a real-time image or video or as thumbnails or icons in interfaces dedicated for presented identifiers of augmented reality experiences). Once an augmented reality experience is selected, one or more images, videos, or augmented reality graphical elements are

retrieved and presented as an overlay on top of the images or video captured by the client device **102**. In some cases, the camera is switched to a front-facing view (e.g., the front-facing camera of the client device **102** is activated in response to activation of a particular augmented reality experience) and the images from the front-facing camera of the client device **102** start being displayed on the client device **102** instead of the rear-facing camera of the client device **102**. The one or more images, videos, or augmented reality graphical elements are retrieved and presented as an overlay on top of the images that are captured and displayed by the front-facing camera of the client device **102**.

(43) In other examples, the augmentation system **208** is implemented as part of an AR headset or AR glasses. In such cases, one or more images are displayed on a transparent display to appear overlaid over real-world objects that are seen by a user wearing the AR headset or AR glasses through the lenses of the AR headset or AR glasses.

(44) In some examples, the augmentation system **208** is able to communicate and exchange data with another augmentation system **208** on another client device **102** and with the server via the network **112**. The data exchanged can include a session identifier that identifies the shared AR session, a transformation between a first client device **102** and a second client device **102** (e.g., a plurality of client devices **102** include the first and second devices) that is used to align the shared AR session to a common point of origin, a common coordinate frame, functions (e.g., commands to invoke functions) as well as other payload data (e.g., text, audio, video or other multimedia data).

(45) The augmentation system **208** sends the transformation to the second client device **102** so that the second client device **102** can adjust the AR coordinate system based on the transformation. In this way, the first and second client devices **102** synchronize their coordinate systems and frames for displaying content in the AR session. Specifically, the augmentation system **208** computes the point of origin of the second client device **102** in the coordinate system of the first client device **102**. The augmentation system **208** can then determine an offset in the coordinate system of the second client device **102** based on the position of the point of origin from the perspective of the second client device **102** in the coordinate system of the second client device **102**. This offset is used to generate the transformation so that the second client device **102** generates AR content according to a common coordinate system or frame as the first client device **102**.

(46) The augmentation system **208** can communicate with the client device **102** to establish individual or shared AR sessions. The augmentation system **208** can also be coupled to the messaging server **118** to establish an electronic group communication session (e.g., group chat, instant messaging) for the client devices **102** in a shared AR session. The electronic group communication session can be associated with a session identifier provided by the client devices **102** to gain access to the electronic group communication session and to the shared AR session. In one example, the client devices **102** first gain access to the electronic group communication session and then obtain the session identifier in the electronic group communication session that allows the client devices **102** to access to the shared AR session. In some examples, the client devices **102** are able to access the shared AR session without aid or communication with the augmentation system **208** in the application servers **114**.

(47) The map system **210** provides various geographic location functions, and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **210** enables the display of user icons or avatars (e.g., stored in profile data **316**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this

location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

(48) The game system **212** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games (e.g., web-based games or web-based applications) that can be launched by a user within the context of the messaging client **104**, and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

(49) The external resource system **220** provides an interface for the messaging client **104** to communicate with external app(s) servers **110** to launch or access external resources. Each external resource (apps) server **110** hosts, for example, a markup language (e.g., HTML5) based application or small-scale version of an external application (e.g., game, utility, payment, or ride-sharing application that is external to the messaging client **104**). The messaging client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the external resource (apps) servers **110** associated with the web-based resource. In certain examples, applications hosted by external resource servers **110** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the messaging server **118**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. In certain examples, the messaging server **118** includes a JavaScript library that provides a given third-party resource access to certain user data of the messaging client **104**. HTML5 is used as an example technology for programming games, but applications and resources programmed based on other technologies can be used.

(50) In order to integrate the functions of the SDK into the web-based resource, the SDK is downloaded by an external resource (apps) server **110** from the messaging server **118** or is otherwise received by the external resource (apps) server **110**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the messaging client **104** into the web-based resource.

(51) The SDK stored on the messaging server **118** effectively provides the bridge between an external resource (e.g., third-party or external applications **109** or applets and the messaging client **104**). This provides the user with a seamless experience of communicating with other users on the messaging client **104**, while also preserving the look and feel of the messaging client **104**. To bridge communications between an external resource and a messaging client **104**, in certain examples, the SDK facilitates communication between external resource servers **110** and the messaging client **104**. In certain examples, a WebViewJavaScriptBridge running on a client device **102** establishes two one-way communication channels between an external resource and the messaging client **104**. Messages are sent between the external resource and the messaging client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

(52) By using the SDK, not all information from the messaging client **104** is shared with external resource servers **110**. The SDK limits which information is shared based on the needs of the external resource. In certain examples, each external resource server **110** provides an HTML5 file corresponding to the web-based external resource to the messaging server **118**. The messaging server **118** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the messaging client **104**. Once the user selects the visual representation or instructs the messaging client **104** through a GUI of the messaging client **104** to access features of

the web-based external resource, the messaging client **104** obtains the HTML5 file and instantiates the resources necessary to access the features of the web-based external resource.

(53) The messaging client **104** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the messaging client **104** determines whether the launched external resource has been previously authorized to access user data of the messaging client **104**. In response to determining that the launched external resource has been previously authorized to access user data of the messaging client **104**, the messaging client **104** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the messaging client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the messaging client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle of or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the messaging client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the messaging client **104**. In some examples, the external resource is authorized by the messaging client **104** to access the user data in accordance with an OAuth 2 framework.

(54) The messaging client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale external applications (e.g., a third-party or external application **109**) are provided with access to a first type of user data (e.g., only two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of external applications (e.g., web-based versions of third-party applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

(55) An AR recommendation system **224** receives an image or video from a client device **102** that depicts a real-world environment (e.g., a room in a home) that includes one or more real-world objects (chair, sofa, television, table, and so forth). The AR recommendation system **224** detects one or more real-world objects depicted in the image or video and uses the detected one or more real-world objects (or features of the real-world environment) to compute a classification for the real-world environment. For example, the AR recommendation system **224** can classify the real-world environment as a kitchen, a bedroom, a nursery, a toddler room, a teenager room, an office, a living room, a den, a formal living room, a patio, a deck, a balcony, a bathroom, or any other suitable real-world environment classification.

(56) Once classified, the AR recommendation system **224** identifies one or more items (such as physical products or electronically consumable content items) related to the real-world environment classification. The identified one or more items can be items that are available for purchase. The AR recommendation system **224** retrieves AR representations (AR items) of the identified items and incorporates (displays at specified positions) the AR representations within the image or video. The AR representations (or AR items) can be interactive, such that upon receiving a user selection or input that selects the particular AR representation, an electronic commerce (e-commerce) purchase transaction is performed to obtain access to or receive the corresponding item.

(57) In some examples, the AR recommendation system **224** modifies pixels of a given real-world object to blend the given real-world object with a background. In this way, the given real-world object can be blended out or removed from the image or video. The AR recommendation system **224** can then place the AR item on the region that has been blended out (e.g., the AR item can be

placed on top of where the real-world object was positioned without overlapping or interfering with the real-world object). This allows the user to see how the AR item looks in the real-world environment as a replacement for the real-world object. In some cases, the AR recommendation system **224** can receive input that drags or moves the AR item to different placements or positions. As the AR item is moved around, the AR recommendation system **224** continuously modifies pixels of real-world objects over which the AR item is placed.
(58) In some cases, the AR recommendation system **224** determines a type of real-world object over which the AR item is placed and determines whether or not to remove or blend the real-world object with the background based on the type of the real-world object. For example, if the real-world object is a coffee machine and the AR item is another kitchen appliance (e.g., a coffee machine or blender), the AR recommendation system **224** can blend pixels of the real-world object with the background to make it appear as though the real-world object has been removed. As another example, if the real-world object is a table and the AR item is a kitchen appliance, the AR recommendation system **224** can determine not to modify pixels of the table because the AR item can be placed on top of the table. In some cases, the AR recommendation system **224** only blends pixels of a real-world object that is of the same or similar type as a type (household appliance of a particular size, such as small, medium or large) of the AR item. An illustrative implementation of the AR recommendation system **224** is shown and described in connection with FIG. **5** below.
(59) The AR recommendation system **224** is a component that can be accessed by an AR/VR application implemented on the client device **102**. The AR/VR application uses an RGB camera to capture an image of a room in a home. The AR/VR application applies various trained machine learning techniques to the captured image of the room to classify the real-world environment. The AR/VR application includes a depth sensor to generate a virtual mesh of the room that is captured in order to incorporate or place into the image or video the AR representations. For example, the AR/VR application can add an AR piece of furniture, such as an AR chair or sofa, to the image or video that is captured by the client device **102**. In some implementations, the AR/VR application continuously captures images of the house or home in real-time or periodically to continuously or periodically update the AR representations of items available for purchase. This allows the user to move around in the real world and see updated AR representations of items available for purchase corresponding to a current real-world environment depicted in an image or video in real-time.
(60) Data Architecture
(61) FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the messaging server system **108**, according to certain examples. While the content of the database **126** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).
(62) The database **126** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **302**, are described below with reference to FIG. **4**.
(63) An entity table **306** stores entity data, and is linked (e.g., referentially) to an entity graph **308** and profile data **316**. Entities for which records are maintained within the entity table **306** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).
(64) The entity graph **308** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.
(65) The profile data **316** stores multiple types of profile data about a particular entity. The profile

data **316** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **316** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100**, and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

(66) Where the entity is a group, the profile data **316** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

(67) The database **126** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **304**) and images (for which data is stored in an image table **312**).

(68) The database **126** can also store data pertaining to individual and shared AR sessions. This data can include data communicated between an AR session client controller of a first client device **102** and another AR session client controller of a second client device **102**, and data communicated between the AR session client controller and the augmentation system **208**. Data can include data used to establish the common coordinate frame of the shared AR scene, the transformation between the devices, the session identifier, images depicting a body, skeletal joint positions, wrist joint positions, feet, and so forth.

(69) Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

(70) Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102**, or the current time.

(71) Other augmentation data that may be stored within the image table **312** includes augmented reality content items (e.g., corresponding to applying augmented reality experiences). An augmented reality content item or augmented reality item may be a real-time special effect and sound that may be added to an image or a video.

(72) As described above, augmentation data includes augmented reality content items, overlays, image transformations, AR images, and similar terms that refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device **102** and then displayed on a screen of the client device **102** with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device **102** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. For example, multiple augmented reality content items that apply different pseudorandom movement models can be applied to the same content by selecting different augmented reality content items for the content. Similarly, real-

time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device **102** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.
(73) Data and various systems using augmented reality content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two-dimensional or three-dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.
(74) Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.
(75) In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of an object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such a method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.
(76) In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the

areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

(77) In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

(78) Other methods and algorithms suitable for face detection can be used. For example, in some examples, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

(79) In some examples, a search is started for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

(80) A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

(81) In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a messaging client **104** operating on the client device **102**. The transformation system operating within the messaging client **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transformation system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user

interface displayed on the client device **102** as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

(82) The graphical user interface, presenting the modification performed by the transformation system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browsing to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

(83) A story table **314** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **306**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

(84) A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client **104**, based on his or her location. The end result is a “live story” told from a community perspective.

(85) A further type of content collection is known as a “location story,” which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

(86) As mentioned above, the video table **304** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. Similarly, the image table **312** stores image data associated with messages for which message data is stored in the entity table **306**. The entity table **306** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **312** and the video table **304**.

(87) The data structures **300** can also store training data for training one or more machine learning techniques (models) to classify a real-world environment. The training data can include a plurality of images and videos and their corresponding ground-truth real-world environment classifications. The ground-truth real-world environment classifications (or any other use of the word ground-truth) refers to the real and correct tag or label that is added to the training data to define the correct

classification or result associated with the training data. The images and videos can include a mix of all sorts of real-world objects that can appear in different real-world environments (e.g., rooms in a home or household). The one or more machine learning techniques can be trained to extract features of a received input image or video and establish a relationship between the extracted features and a real-world environment classification. Once trained, the machine learning technique can receive a new image or video and can compute a real-world environment classification for the newly received image or video.
(88) The data structures **300** can also store training data for training one or more machine learning techniques (models) to determine a blending pattern for an object. The training data can include a plurality of images and videos, real-world object labels of real-world objects that appear in the images and videos, and their corresponding ground-truth blending patterns. The images and videos can include a mix of all sorts of real-world objects that can appear in different real-world environments (e.g., rooms in a home or household). The one or more machine learning techniques can be trained to extract features of a received input image or video and establish a relationship between the extracted features, the objects detected in the image or video, and a blending pattern for each object. Once trained, the machine learning technique can receive a new image or video, a given real-world object, and can compute a blending pattern for the real-world object that appears in the newly received image or video.
(89) The data structures **300** can also store a list or plurality of different expected objects for different real-world environment classifications. For example, the data structures **300** can store a first list of expected objects for a first real-world environment classification. Namely, a real-world environment classified as a kitchen can be associated with a list of expected objects, such as appliances and/or furniture items including: tea maker, toaster, kettle, mixer, refrigerator, blender, cabinet, cupboard, cooker hood, range hood, microwave, dish soap, kitchen counter, dinner table, kitchen scale, pedal bin, grill, and drawer. As another example, a real-world environment classified as a living room can be associated with a list of expected objects including: wing chair, TV stand, sofa, cushion, telephone, television, speaker, end table, tea set, fireplace, remote, fan, floor lamp, carpet, table, blinds, curtains, picture, vase, and grandfather clock.
(90) As another example, a real-world environment classified as a bedroom can be associated with a list of expected objects, such as furniture items including: headboard, footboard and mattress frame, mattress and box springs, mattress pad, sheets and pillowcases, blankets, quilts, comforter, bedspread, duvet, bedskirt, sleeping pillows, specialty pillows, decorative pillows, pillow covers and shams, throws (blankets), draperies, rods, brackets, valances, window shades, blinds, shutters, nightstands, occasional tables; lamps; floor, table, hanging; wall sconces, alarm clock, radio, plants and plant containers, vases, flowers, candles, candleholders, artwork, posters, prints, photos, frames, photo albums, decorative objects and knick-knacks, dressers and clothing, armoire, closet, TV cabinet, chairs, loveseat, chaise lounge, ottoman, bookshelves, decorative ledges, books, magazines, bookends, trunk, bench, writing desk, vanity table, mirrors, rugs, jewelry boxes and jewelry, storage boxes, baskets, trays, telephone; television, cable box, satellite box, DVD player and videos, tablets, nightlight.
(91) Each of the expected objects can be associated with a label identifying a type of the expected object and a range of sizes associated with the expected object. The label can then be used to selectively modify pixels (e.g., apply a blending pattern to remove the expected object from the image or video) of the expected object (if the expected object appears in the real-world environment) in response to an AR item being placed on top of the expected object in the real-world environment.
(92) Data Communications Architecture
(93) FIG. **4** is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by a messaging client **104** for communication to a further messaging client **104** or the messaging server **118**. The content of a particular message **400** is used to populate the

message table **302** stored within the database **126**, accessible by the messaging server **118**. Similarly, the content of a message **400** is stored in memory as “in-transit” or “in-flight” data of the client device **102** or the application servers **114**. A message **400** is shown to include the following example components: message identifier **402**: a unique identifier that identifies the message **400**. message text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the message **400**. message image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **312**. message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the video table **304**. message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the message **400**. message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data **412** for a sent or received message **400** may be stored in the augmentation table **310**. message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the messaging client **104**. message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **406**, or a specific video in the message video payload **408**). message story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **314**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple images within the message image payload **406** may each be associated with multiple content collections using identifier values. message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition. message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** on which the message **400** was generated and from which the message **400** was sent. message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the message **400** is addressed.

(94) The contents (e.g., values) of the various components of message **400** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **312**. Similarly, values within the message video payload **408** may point to data stored within a video table **304**, values stored within the message augmentation data **412** may point to data stored in an augmentation table **310**, values stored within the message story identifier **418** may point to data stored in a story table **314**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **306**.

(95) AR Recommendation System

(96) FIG. **5** is a block diagram showing an example AR recommendation system **224**, according to example examples. The AR recommendation system **224** includes a set of components **510** that

operate on a set of input data (e.g., a monocular image (or video) depicting a real-world physical environment **501** (e.g., a room in a home) and depth map data **502** (e.g., depth data received from a LiDAR sensor of the client device **102**). The AR recommendation system **224** includes an object detection module **512**, a room classification module **514**, a depth reconstruction module **517**, an expected object module **516**, an image modification module **518**, an AR item selection module **519**, and an image display module **520**. All or some of the components of the AR recommendation system **224** can be implemented by a server, in which case, the monocular image depicting a real-world physical environment **501** (e.g., a room in a home) and the depth map data **502** are provided to the server by the client device **102**. In some cases, some or all of the components of the AR recommendation system **224** can be implemented by the client device **102**.

(97) The object detection module **512** receives a monocular image (or video) depicting a real-world physical environment **501**. This image can be received as part of a real-time video stream, a previously captured video stream or a new image captured by a (front-facing or rear-facing) camera of the client device **102**. The object detection module **512** applies one or more machine learning techniques to identify real-world physical objects that appear in the image depicting a real-world physical environment **501**. For example, the object detection module **512** can segment out individual objects in the image and assign a label or name to the individual objects. Specifically, the object detection module **512** can recognize a sofa as an individual object, a television as another individual object, a light fixture as another individual object, and so forth. Any type of object that can appear or be present in a particular home or household can be recognized and labeled by the object detection module **512**. The label can specify a type of the detected object (e.g., one or more furniture types, one or more appliance types, one or more household item types) and a size of the specified type (e.g., small, medium or large).

(98) As another example, the object detection module **512** extracts features from the detected objects and provides such features as input to a trained machine learning model (e.g., neural network). The trained machine learning model can then use the input to predict a proper classification for each of the detected objects. In particular, the machine learning model can determine probability values for a set of predetermined classifiers that indicate a likelihood that each classifier properly classifies the detected objects. The object detection module **512** can then select the classifier with the highest probability value as the identified real-world physical objects that appear in the image depicting a real-world physical environment **501**.

(99) The object detection module **512** provides the identified and recognized objects to the room classification module **514** (e.g., a real-world environment classification module). The room classification module **514** can compute or determine a real-world environment classification of the real-world environment depicted in the image or video depicting the real-world physical environment **501** based on the identified and recognized objects received from the object detection module **512**. In some implementations, the room classification module **514** compares the objects received from the object detection module **512** to a plurality of lists of expected objects each associated with a different real-world environment classification that is stored in data structures **300**. For example, the room classification module **514** can compare the objects detected by the object detection module **512** to a first list of expected objects associated with a living room classification. The room classification module **514** can compute a quantity or percentage of the objects that are detected by the object detection module **512** and that are included in the first list. The room classification module **514** can assign a relevancy score to the first list. The room classification module **514** can then similarly compare the objects detected by the object detection module **512** to a second list of expected objects associated with another room classification (e.g., a kitchen). The room classification module **514** can then compute a quantity or percentage of the objects that are detected by the object detection module **512** and that are included in the second list and can assign a relevancy score to the second list. The room classification module **514** can identify which of the lists that are stored in the data structures **300** is associated with a highest

relevancy score. The room classification module **514** can then determine or compute the real-world environment classification of the real-world environment depicted in the image or video depicting the real-world physical environment **501** based on the room classification associated with the identified list of expected objects with the highest relevancy score.

(100) In another implementation, the room classification module **514** can implement one or more machine learning techniques to classify a real-world environment (e.g., a room in a home or household). The machine learning techniques can implement a classifier neural network that is trained to establish a relationship between one or more features of an image or video of a real-world environment with a corresponding real-world environment classification.

(101) During training, the machine learning technique of the room classification module **514** receives a given training image (e.g., a monocular image or video depicting a real-world environment, such as an image of a living room or bedroom) from training image data stored in data structures **300**. The room classification module **514** applies one or more machine learning techniques to the given training image. The room classification module **514** extracts one or more features from the given training image to estimate a real-world environment classification for the real-world environment depicted in the image or video. For example, the room classification module **514** obtains the given training image depicting a real-world environment and extracts features from the image or video that correspond to the real-world objects that appear in the real-world environment. In some cases, rather than receiving an image or video depicting a real-world environment, the room classification module **514** receives a list or plurality of objects detected by another module or machine learning technique. The room classification module **514** is trained to determine a real-world environment classification based on the features of the objects received from the other machine learning technique.

(102) The room classification module **514** determines the relative positions of the detected real-world objects and/or features of the image or video depicting the real-world environment. The room classification module **514** then estimates or computes a real-world environment classification based on the relative positions of the detected real-world objects and/or features of the image depicting the real-world environment. The room classification module **514** obtains a known or predetermined ground-truth room classification of the real-world environment depicted in the training image from the training data. The room classification module **514** compares the estimated real-world environment classification with the ground truth room classification. Based on a difference threshold of the comparison, the room classification module **514** updates one or more coefficients or parameters and obtains one or more additional training images of a real-world environment. In some cases, the room classification module **514** is first trained on a set of images associated with one real-world environment classification and is then trained on another set of images associated with another real-world environment classification.

(103) After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (computed as a function of a difference between the estimated classification and the ground-truth classification) reaches a specified value, the room classification module **514** completes training and the parameters and coefficients of the room classification module **514** are stored as a trained machine learning technique or trained classifier.

(104) In some cases, multiple classifiers are trained in parallel or sequentially on different sets of training data corresponding to different real-world environment classifications. For example, a first classifier can be trained to classify a living room based on a first set of training images that depict different living room features. A second classifier can be trained to classify a bedroom based on a second set of training images that depict different bedroom features. The bedroom classifier can also provide an output that provides an estimated age of the person associated with the bedroom. In this way, the bedroom classifier can indicate that the bedroom is a master or guest bedroom (if the estimated age is within a first range), a nursery (if the estimated age is within a second range of ages that are younger than the ages of the first range), a toddler room (if the estimated age range is

within a third range), or a teenager room (if the estimated age range is within a fourth range). In such circumstances, multiple classifiers can operate on a same input image and can generate a real-world environment classification with a given score that indicates how accurate the generated real-world environment classification is. The room classification module **514** obtains the scores and classifications from all of the multiple classifiers and then assigns the real-world environment classification to the input monocular image depicting the real-world physical environment **501** based on the real-world environment classification with the highest score.
(105) In an example, after training, the room classification module **514** receives an input monocular image depicting a real-world physical environment **501** as a single RGB image from a client device **102** or as a video of multiple images. The room classification module **514** applies the trained machine learning technique(s) to the received input image to extract one or more features and to generate a prediction or estimation of the real-world environment classification of the image depicting a real-world physical environment **501**.
(106) In some examples, the room classification module **514** is trained to estimate a blending pattern for each real-world object that appears in a real-world environment. The blending pattern is used to blend in pixels of a real-world object with a background of the real-world environment to make it appear as though the real-world object has been removed. Different real-world objects can be associated with different blending patterns.
(107) During training, the machine learning technique of the room classification module **514** receives a given training image (e.g., a monocular image or video depicting a real-world environment, such as an image of a living room or bedroom) from training image data stored in data structures **300** along with a list of real-world objects that appear in the real-world environment. The room classification module **514** applies one or more machine learning techniques to the given training image. The room classification module **514** extracts one or more features from the given training image and the list of real-world objects to estimate a blending pattern for each of the real-world objects. For example, the room classification module **514** obtains the given training image depicting a real-world environment and a first set of real-world objects and extracts features from the image or video that correspond to the real-world objects that appear in the real-world environment.
(108) The room classification module **514** then estimates or computes a blending pattern associated with each of the detected real-world objects and/or features of the image depicting the real-world environment. The room classification module **514** obtains a known or predetermined ground-truth blending pattern of each real-world object depicted in the training image from the training data. The room classification module **514** compares the estimated blending pattern with the ground truth room blending pattern. Based on a difference threshold of the comparison, the room classification module **514** updates one or more coefficients or parameters and obtains one or more additional training images of a real-world environment. In some cases, the room classification module **514** is first trained on a set of images associated with one real-world environment and is then trained on another set of images associated with another real-world environment.
(109) After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (computed as a function of a difference between the estimated classification and the ground-truth classification) reaches a specified value, the room classification module **514** completes training and the parameters and coefficients of the room classification module **514** are stored as a trained machine learning technique or trained classifier.
(110) In an example, the room classification module **514** provides the room classification to the expected object module **516**. The expected object module **516** obtains a list or plurality of expected objects stored in the data structures **300** that is associated with the room classification. For example, the expected object module **516** obtains list of expected objects (furniture items) including: headboard, footboard and mattress frame, mattress and box springs, mattress pad, sheets and pillowcases, blankets, quilts, comforter, bedspread, duvet, bedskirt, sleeping pillows, specialty

pillows, decorative pillows, pillow covers and shams, throws (blankets), draperies, rods, brackets, valances, window shades, blinds, shutters, nightstands, occasional tables; lamps: floor, table, hanging; wall sconces, alarm clock, radio, plants and plant containers, vases, flowers, candles, candleholders, artwork, posters, prints, photos, frames, photo albums, decorative objects and knick-knacks, dressers and clothing, armoire, closet, TV cabinet, chairs, loveseat, chaise lounge, ottoman, bookshelves, decorative ledges, books, magazines, bookends, trunk, bench, writing desk, vanity table, mirrors, rugs, jewelry boxes and jewelry, storage boxes, baskets, trays, telephone; television, cable box, satellite box, DVD player and videos, tablets, nightlight.

(111) The expected object module **516** compares the listed items from the retrieved list of expected objects with the objects detected by the object detection module **512** and/or with the objects identified by the machine learning technique implemented by the room classification module **514**. The expected object module **516** can determine that a particular expected object is missing from the list of detected objects. For example, the expected object module **516** can determine that the room classification is a bedroom and that a bed furniture item is missing from the list of objects provided by the object detection module **512** or room classification module **514**. In such cases, the expected object module **516** can select a bed as a furniture item to recommend to a user to purchase for inclusion in the room depicted in the image captured by the client device **102**. Specifically, the expected object module **516** provides the identifier of the bed furniture item to the AR item selection module **519**. The AR item selection module **519** can then search for and retrieve an AR representation of the bed furniture item from a list of AR representations of beds and can instruct the image modification module **518** to incorporate the AR representation of the bed furniture item into the image captured by the client device **102**. In some cases, the AR item selection module **519** provides an indication of where to place the AR representation relative to one or more other real-world objects depicted in the image. After the AR representation is incorporated into the image, the image display module **520** generates for display an image that includes the AR representation of the bed furniture item together with other real-world objects (real-world furniture items) depicted in the image captured by the client device **102**.

(112) As another example, the expected object module **516** can determine that a particular expected object is missing from the list of detected objects, such as an appliance is missing from the kitchen that is depicted in the image. For example, the expected object module **516** can determine that the room classification is a kitchen (e.g., in response to detecting a sink and stove among objects depicted in a video) and that a mixer kitchen appliance is missing from the list of objects provided by the object detection module **512** or room classification module **514**. In such cases, the expected object module **516** can select a mixer appliance as an appliance item to recommend to a user to purchase for inclusion in the room depicted in the image captured by the client device **102**. Specifically, the expected object module **516** provides the identifier of the mixer appliance to the AR item selection module **519**. The AR item selection module **519** can then search for and retrieve an AR representation of the mixer appliance from a list of AR representations of mixer appliances and can instruct the image modification module **518** to incorporate the AR representation of the mixer appliance into the image captured by the client device **102**. In some cases, the AR item selection module **519** provides an indication of where to place the AR representation relative to one or more other real-world objects depicted in the image. After the AR representation is incorporated into the image, the image display module **520** generates for display an image that includes the AR representation of the mixer appliance together with other real-world objects (real-world furniture items) depicted in the image captured by the client device **102**. The client device **102** can receive a user input that selects the AR representation and in response completes a purchase transaction for the mixer appliance associated with the AR representation.

(113) As another example, the expected object module **516** can determine that a particular expected object matches a given object from the list of detected objects, such as an appliance in the kitchen. For example, the expected object module **516** can detect a coffee machine appliance in the list of

detected objects. The expected object module **516** can estimate a make and model of the coffee machine appliance and can select another coffee machine appliance (e.g., another kitchen appliance of a similar type as the detected kitchen appliance) from the expected object list. The selected coffee machine appliance can correspond to a newer model coffee machine as compared to the make and model of the detected coffee machine appliance in the list of detected objects. The expected object module **516** can instruct the AR item selection module **519** to obtain an AR item or AR representation of the selected coffee machine appliance.

(114) The depth reconstruction module **517** receives depth map data **502** from a depth sensor or depth camera of the client device **102**. The depth map data **502** is associated with the image or video being processed by the room classification module **514** and the expected object module **516**. The depth reconstruction module **517** can generate a three-dimensional (3D) mesh representation, a 3D model, or reconstruction of the room depicted in the image captured by the client device **102**. The depth reconstruction module **517** can provide the 3D mesh representation or reconstruction of the room or 3D model to the expected object module **516**. The expected object module **516** can search through the list of expected objects that can include multiple versions or types of the selected coffee machine appliance each having a different size and configuration. The expected object module **516** can compute a size of each of the coffee machine appliances in the list and determine, based on the 3D model, a given one of the coffee machine appliances that fits within the region of the 3D model in which the detected coffee machine appliance is located.

(115) Specifically, based on the 3D mesh representation or reconstruction, the expected object module **516** can further refine which objects from the expected object list to recommend to the user to purchase. The expected object module **516** can process the 3D mesh representation of the room to compute an amount of available physical space remaining in the real-world environment depicted in the image. For example, based on the 3D mesh representation or reconstruction, the expected object module **516** can further refine which objects from the expected object list to recommend to the user to purchase that fit better within the physical space of the room.

(116) The AR item selection module **519** selects a given AR item or AR representation of a coffee machine that is determined to fit best among all the available coffee machines in the expected item list. The AR item selection module **519** can then obtain a blending pattern associated with the detected coffee machine appliance that is in the image or video. For example, the AR item selection module **519** can communicate the identification of the detected coffee machine appliance to the room classification module **514**. The room classification module **514** obtains the image or video that depicts the detected coffee machine appliance and applies a machine learning technique to the image or video and the detected coffee machine appliance. The AR item selection module **519** estimates a blending pattern for the coffee machine appliance based on an output of the machine learning technique.

(117) The AR item selection module **519** provides the identification of the coffee machine appliance, the blending pattern, and the AR representation or AR item to the image modification module **518**. The image modification module **518** determines that the AR representation or AR item has been placed in a position in the image or video that overlaps the coffee machine appliance (e.g., the real-world object depicted in the image or video). In response, the image modification module **518** computes a type and size associated with the real-world object. In this case, the type is determined to be a kitchen appliance and the size is determined to be small. The image modification module **518** also obtains a type and size of the AR representation or AR item (e.g., kitchen appliance and small). The image modification module **518** also determines that the type and size of the real-world object matches the type and size of the AR representation or AR item. In response, the image modification module **518** applies the blending pattern to a region of the image or video that includes the real-world object.

(118) For example, the image modification module **518** modifies pixels of the real-world object to blend the real-world object with a background of the real-world environment depicted in the image

or video. As a result, the real-world object is removed from the image or video and is replaced with a blurred region. The image modification module **518** then displays the AR representation or AR item on the blurred region to make it appear as though the AR representation or AR item has been placed in the real-world environment in replacement of the real-world object. Namely, the image modification module **518** modifies pixels so that the real-world coffee machine appliance appears to be replaced with the AR coffee machine appliance.

(119) The image modification module **518** also determines an orientation of a surface on which the AR item or representation is placed. The image modification module **518** modifies the orientation of the AR item or representation to match the orientation of the surface on which the AR item or representation is placed.

(120) The image modification module **518** can receive input from a user that drags or moves the AR item or representation to a new position or placement in the image or video. In response to determining that the AR item or representation no longer overlaps the real-world object that has had the pixels modified, the image modification module **518** undoes the modification to the pixels and returns the pixel values to the original value in the image or video. Namely, the image modification module **518** returns into view the real-world object (e.g., the coffee machine appliance) when the AR item or representation is moved away from a position that overlaps the real-world object. The image modification module **518** determines that the AR representation or AR item has been placed in a new position in the image or video that overlaps a different real-world object (e.g., a cabinet). In response, the image modification module **518** computes a type and size associated with the real-world object. In this case, the type is determined to be a furniture item and the size is determined to be medium. The image modification module **518** also obtains a type and size of the AR representation or AR item (e.g., kitchen appliance and small). The image modification module **518** also determines that the type and size of the real-world object fail to match the type and size of the AR representation or AR item. In response, the image modification module **518** does not apply the blending pattern to a region of the image or video that includes the real-world object and allows the AR representation or item to overlap the real-world object. Namely, the image modification module **518** displays the AR representation on top of the real-world object without removing the real-world object from the display. In some cases, the image modification module **518** modifies the orientation of the AR representation or item to match an orientation of the real-world object.

(121) The image modification module **518** can receive input from a user that drags or moves the AR item or representation to a new position or placement in the image or video. In response to determining that the AR item or representation no longer overlaps the real-world object that has had the pixels modified, the image modification module **518** undoes the modification to the pixels and returns the pixel values to the original value in the image or video. Namely, the image modification module **518** returns into view the real-world object (e.g., the coffee machine appliance) when the AR item or representation is moved away from a position that overlaps the real-world object. The image modification module **518** determines that the AR representation or AR item has been placed in a new position in the image or video that overlaps a different real-world object (e.g., a blender appliance). In response, the image modification module **518** computes a type and size associated with the real-world object. In this case, the type is determined to be a kitchen appliance and the size is determined to be small. The image modification module **518** also obtains a type and size of the AR representation or AR item (e.g., kitchen appliance and small). The image modification module **518** also determines that the type and size of the real-world object match the type and size of the AR representation or AR item. In response, the image modification module **518** communicates with the room classification module **514** to obtain a blending pattern for the blender kitchen appliance. The image modification module **518** can determine that the blending pattern is a second blending pattern that differs from the blending pattern associated with the coffee kitchen appliance. In such cases, the image modification module **518** applies the blending pattern to

a region of the image or video that includes the blender kitchen appliance and allows the AR representation or item to overlap the blender kitchen appliance. Namely, the image modification module **518** displays the AR representation on top of the region where the blender kitchen appliance was located and removed from the display.

(122) In some cases, the image modification module **518** determines an orientation of the real-world object over which the AR representation or item is placed. Namely, after or before modifying the pixels of the real-world object to remove the real-world object from the display, the image modification module **518** can apply a ray-tracing process to determine an orientation of the real-world object. The image modification module **518** then adjusts the orientation of the AR representation or item to match the orientation of the real-world object. The image modification module **518** then places the AR representation or item on the region of the real-world object that has been blurred out (where the real-world object was removed) in the same orientation as the real-world object. This provides the illusion that the AR representation or item is part of the real-world environment.

(123) FIGS. **6-8** are diagrammatic representations of outputs of the AR recommendation system **224**, in accordance with some examples. Specifically, as shown in the user interface **600** of FIG. **6**, the AR recommendation system **224** receives an image or video **610** that depicts a real-world environment. The AR recommendation system **224** applies one or more machine learning techniques to detect and recognize one or more real-world objects that are depicted in the image or video **610**. For example, the AR recommendation system **224** detects and recognizes a coffee machine appliance **620**, among other objects.

(124) Based on the detected and recognized objects, the AR recommendation system **224** determines that the real-world environment depicted in the image or video **610** corresponds to a kitchen classification. In response, the AR recommendation system **224** obtains a list of expected objects associated with the kitchen classification. The AR recommendation system **224** determines that the coffee machine appliance **620** matches a given one of the objects in the list of expected objects. In response, the AR recommendation system **224** obtains size information for the coffee machine appliance **620** and selects an AR item that matches the size information of the coffee machine appliance **620** from the list of expected objects.

(125) The AR recommendation system **224** provides the image or video **610** and the detected real-world object (e.g., the coffee machine appliance **620**) to a trained machine learning technique. The machine learning technique provides a blending pattern for the detected real-world object. The AR recommendation system **224** applies the blending pattern to the coffee machine appliance **620** to modify pixels of the coffee machine appliance **620**. For example, as shown in the user interface **700** of FIG. **7**, the AR recommendation system **224** modifies the pixels of the coffee machine appliance to generate an image or video **710** with a blurred region **720**. The blurred region **720** blends pixel values of the coffee machine appliance **620** with the background. For example, the blurred region **720** blends pixels values of the coffee machine appliance **620** with adjacent pixels values. This makes it appear as though the coffee machine appliance **620** has been removed from the image or video **710**.

(126) After removing the coffee machine appliance **620** from the image or video, the AR recommendation system **224** adds the AR item (corresponding to an AR coffee machine appliance) to the real-world environment depicted in the image or video. For example, as shown in the user interface **800** of FIG. **8**, the AR item **830** has been added on top of the blurred region **820** (where the real-world object was removed from the image or video **810**). This makes it appear as though the AR item **830** is part of the real-world environment and has replaced the real-world object (e.g., the coffee machine appliance **620**). The AR recommendation system **224** can modify an orientation of the AR item **830** to match the orientation of a surface on which the AR item **830** is placed and/or the orientation of the real-world object that has been removed (e.g., the orientation of the coffee machine appliance **620**).

(127) In some cases, the AR recommendation system **224** can receive input that moves the AR item **830** to a new position that no longer overlaps the real-world object (e.g., the coffee machine appliance **620**). The AR recommendation system **224** can, in response, redisplay (undo the blurring of the pixels of the coffee machine appliance **620**) and can selectively blur out or remove pixels of another real-world object that is overlapped by the AR item **830**. For example, if the AR recommendation system **224** determines that that AR item **830** has been moved to a new position that overlaps a first type of real-world object, the AR recommendation system **224** can allow the AR item **830** to be displayed together with the first type of real-world object (e.g., a table). As another example, if the AR recommendation system **224** determines that the AR item **830** has been moved to a new position that overlaps a second type of real-world object (e.g., a type of real-world object that matches the type of AR item **830**), the AR recommendation system **224** can modify pixel values of the second type of real-world object to blur out and remove the second type of real-world object from being displayed.
(128) While the disclosed examples are provided in the context of a coffee machine kitchen appliance being replaced with an AR coffee machine appliance, similar operations can be performed for replacing any other type of real-world object (e.g., television, refrigerator, and so forth) with a corresponding type of AR item (e.g., AR television, AR refrigerator, and so forth).
(129) In some examples, the AR recommendation system **224** receives an image or video that depicts a real-world environment. The AR recommendation system **224** generates a classification of the real-world environment. The AR recommendation system **224** can identify a particular real-world object that appears in the real-world environment. The AR recommendation system **224** can determine a type and size associated with the particular real-world object. The AR recommendation system **224** can retrieve a list of AR items or representations associated with the classification. The AR recommendation system **224** obtains a subset of AR items or representations from the list that includes AR items or representations corresponding to the type and size of the particular real-world object. The AR recommendation system **224** determines attributes (e.g., make and model) of each of the AR items or representations on the subset and selects a given AR item or representation that matches a profile of a user of the client device **102**.
(130) The AR recommendation system **224** obtains a blending parameter of the particular real-world object. The AR recommendation system **224** modifies pixels of the particular real-world object to generate a blurred region where all of the pixels of the particular real-world object are blended into a background of the real-world object. This makes it appear as though the real-world object has been removed from the display. The AR recommendation system **224** then adds the AR item or representation to the same location and position of the blurred region. In an example, the AR recommendation system **224** can modify an orientation of the AR item or representation to match the orientation of the surface on which the AR item or representation is placed. The AR recommendation system **224** can also modify the orientation of the AR item or representation to match the orientation of the particular real-world object.
(131) FIG. **9** is a flowchart of a process **900**, in accordance with some examples. Although the flowchart can describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part, may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.
(132) At operation **901**, a client device **102** receives a video that includes a depiction of a real-world object in a real-world environment, as discussed above.
(133) At operation **902**, the client device **102** determines a classification for the real-world environment by processing the real-world object depicted in the video, as discussed above.

(134) At operation **903**, the client device **102** selects an AR item based on the classification of the real-world environment and the real-world object depicted in the video, as discussed above.
(135) At operation **904**, the client device **102** modifies pixels corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object, as discussed above.
(136) At operation **905**, the client device **102** adds the AR item to the modified video at a display position corresponding to the removed pixels, as discussed above.
(137) Machine Architecture
(138) FIG. **10** is a diagrammatic representation of a machine **1000** within which instructions **1008** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1008** may cause the machine **1000** to execute any one or more of the methods described herein. The instructions **1008** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. The machine **1000** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1008**, sequentially or otherwise, that specify actions to be taken by the machine **1000**. Further, while only a single machine **1000** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1008** to perform any one or more of the methodologies discussed herein. The machine **1000**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1000** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.
(139) The machine **1000** may include processors **1002**, memory **1004**, and input/output (I/O) components **1038**, which may be configured to communicate with each other via a bus **1040**. In an example, the processors **1002** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1006** and a processor **1010** that execute the instructions **1008**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **10** shows multiple processors **1002**, the machine **1000** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.
(140) The memory **1004** includes a main memory **1012**, a static memory **1014**, and a storage unit **1016**, all accessible to the processors **1002** via the bus **1040**. The main memory **1004**, the static memory **1014**, and the storage unit **1016** store the instructions **1008** embodying any one or more of the methodologies or functions described herein. The instructions **1008** may also reside, completely or partially, within the main memory **1012**, within the static memory **1014**, within a machine-

readable medium within the storage unit **1016**, within at least one of the processors **1002** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**.

(141) The I/O components **1038** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific U/O components **1038** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the U/O components **1038** may include many other components that are not shown in FIG. **10**. In various examples, the I/O components **1038** may include user output components **1024** and user input components **1026**. The user output components **1024** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1026** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(142) In further examples, the I/O components **1038** may include biometric components **1028**, motion components **1030**, environmental components **1032**, or position components **1034**, among a wide array of other components. For example, the biometric components **1028** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1030** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(143) The environmental components **1032** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(144) With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 3600 photographs and videos.

(145) Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera

configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.
(146) The position components **1034** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.
(147) Communication may be implemented using a wide variety of technologies. The I/O components **1038** further include communication components **1036** operable to couple the machine **1000** to a network **1020** or devices **1022** via respective coupling or connections. For example, the communication components **1036** may include a network interface component or another suitable device to interface with the network **1020**. In further examples, the communication components **1036** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1022** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).
(148) Moreover, the communication components **1036** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1036** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1036**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.
(149) The various memories (e.g., main memory **1012**, static memory **1014**, and memory of the processors **1002**) and storage unit **1016** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1008**), when executed by processors **1002**, cause various operations to implement the disclosed examples.
(150) The instructions **1008** may be transmitted or received over the network **1020**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1036**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1008** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1022**.
(151) Software Architecture
(152) FIG. **11** is a block diagram **1100** illustrating a software architecture **1104**, which can be installed on any one or more of the devices described herein. The software architecture **1104** is supported by hardware such as a machine **1102** that includes processors **1120**, memory **1126**, and I/O components **1138**. In this example, the software architecture **1104** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1104** includes layers such as an operating system **1112**, libraries **1110**, frameworks **1108**, and applications **1106**. Operationally, the applications **1106** invoke API calls **1150** through the software stack and receive messages **1152** in response to the API calls **1150**.
(153) The operating system **1112** manages hardware resources and provides common services. The operating system **1112** includes, for example, a kernel **1114**, services **1116**, and drivers **1122**. The

kernel **1114** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1114** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **1116** can provide other common services for the other software layers. The drivers **1122** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1122** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(154) The libraries **1110** provide a common low-level infrastructure used by the applications **1106**. The libraries **1110** can include system libraries **1118** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1110** can include API libraries **1124** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1110** can also include a wide variety of other libraries **1128** to provide many other APIs to the applications **1106**.

(155) The frameworks **1108** provide a common high-level infrastructure that is used by the applications **1106**. For example, the frameworks **1108** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **1108** can provide a broad spectrum of other APIs that can be used by the applications **1106**, some of which may be specific to a particular operating system or platform.

(156) In an example, the applications **1106** may include a home application **1136**, a contacts application **1130**, a browser application **1132**, a book reader application **1134**, a location application **1142**, a media application **1144**, a messaging application **1146**, a game application **1148**, and a broad assortment of other applications such as an external application **1140**. The applications **1106** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1106**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the external application **1140** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the external application **1140** can invoke the API calls **1150** provided by the operating system **1112** to facilitate functionality described herein.

(157) Glossary

(158) “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

(159) “Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based

or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

(160) “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(161) “Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions.

(162) Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

(163) A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

(164) Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.
(165) Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).
(166) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1002** or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.
(167) “Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.
(168) “Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(169) “Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”
(170) “Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.
(171) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a matter as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.
(172) Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

Claims

1. A method comprising: receiving, by one or more processors, a video that includes a depiction of a real-world object in a real-world environment; determining, by a trained neural network, a classification for the real-world environment by processing the real-world object depicted in the video; selecting an augmented reality (AR) item based on the classification of the real-world environment and based on the real-world object depicted in the video; modifying pixel data corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object; adding a depiction of the AR item to the modified video at a display position corresponding to the modified pixel data, the adding the depiction of the AR item to the modified video comprising: calculating characteristic points for a set of elements of the real-world object to generate a mesh based on the calculated characteristic points; generating one or more areas on the mesh of the real-world object; aligning a position of the one or more areas of the real-world object with one or more elements of the AR item; and modifying one or more visual properties of the one or more areas to cause a user device to display the AR item within the video at an individual display position relative to the display position of the real-world object that has had pixel data modified; detecting input that moves the AR item to a new position in the video; determining that the AR item in the new position no longer overlaps a portion of the real-world object that has had the corresponding pixel data modified; and in response to determining that the AR item in the new position no longer overlaps the portion of the real-world object that has had the corresponding pixel data modified, undoing modification of the pixel data to return pixel values of the real-world object to an original value in the video.
2. The method of claim 1, further comprising generating, for display, the modified video with the

depiction of the AR item that has been added.

3. The method of claim 1, further comprising blurring a region corresponding to the modified pixel data, wherein the depiction of the AR item is added to the blurred region.

4. The method of claim 3, further comprising blending pixel values in the blurred region with pixel values of other real-world objects that are adjacent to the blurred region.

5. The method of claim 1, further comprising applying a machine learning technique to the video to modify the pixel data and generate the modified video.

6. The method of claim 5, wherein the machine learning technique is trained to establish a relationship between different types of real-world objects and image blending patterns.

7. The method of claim 6, further comprising training the machine learning technique by: receiving training data comprising a plurality of training images and ground truth room blending patterns for each of the plurality of training images, each of the plurality of training images depicting a different real-world environment having different real-world object types; selecting a first real-world object depicted in a first training image of the plurality of training images; applying the neural network to the first training image and the first real-world object to estimate a blending pattern for the real-world environment depicted in the first training image; computing a deviation between the estimated blending pattern and the ground truth room blending pattern associated with the first training image; updating parameters of the neural network based on the computed deviation; and repeating the applying, computing and updating steps for a set of the plurality of training images.

8. The method of claim 1, further comprising: obtaining a plurality of excluded objects associated with the classification; detecting the real-world object depicted in the video using an object recognition process; and comparing the detected object depicted in the video to the plurality of expected excluded objects.

9. The method of claim 8, further comprising: based on the comparing, identifying a given excluded object from the plurality of excluded objects that is excluded from the detected real-world object; and searching for an AR item corresponding to the given excluded object.

10. The method of claim 9, further comprising: generating a three-dimensional (3D) mesh representation of the real-world environment; obtaining a plurality of real-world items corresponding to the classification; detecting that the plurality of real-world items excludes the detected object depicted in the video; determining, based on the 3D mesh representation, that physical space is available for a given one of the plurality of real-world items; and selecting an AR item corresponding to the given one of the plurality of real-world items.

11. The method of claim 1, wherein the classification corresponds to a kitchen, the method further comprising: detecting that the real-world object depicted in the video corresponds to a first type of kitchen appliance; and searching a plurality of AR items to identify an AR item corresponding to the first type of kitchen appliance, wherein the AR item corresponding to the first type of kitchen appliance represents a model of the first type of kitchen appliance that is different than the real-world object.

12. The method of claim 1, wherein determining the classification for the real-world environment comprises: comparing, by the trained neural network, the real-world object depicted in the video to a plurality of lists of expected objects, each list associated with a different real-world environment classification; computing, for each list, a relevancy score based on a quantity or percentage of objects depicted in the video that match objects in the list; identifying the list with the highest relevancy score; and determining the classification for the real-world environment based on the real-world environment classification associated with the identified list having the highest relevancy score.

13. The method of claim 1, further comprising: generating a three-dimensional (3D) mesh representation of the real-world environment; detecting that the real-world object depicted in the video fails to satisfy one or more fit parameters of the 3D mesh representation; identifying, based

on the 3D mesh representation, a recommended item that satisfies the one or more fit parameters of the 3D mesh representation and is of a same type as the real-world object included in the video; and retrieving an AR item corresponding to the recommended item in response to identifying the recommended item.

14. The method of claim 1, wherein determining the classification for the real-world environment further comprises: applying, by the trained neural network, multiple classifiers to the video, wherein at least one classifier is trained to classify a room and provide an estimated age range associated with the room; determining, based on the estimated age range, a specific room type classification; generating, by each classifier, a classification score indicating an accuracy of the generated real-world environment classification; and assigning the real-world environment classification based on the classification with the highest score among the multiple classifiers.

15. The method of claim 14, wherein the specific room type classification is selected from a plurality of different types of bedrooms each associated with a different age range, further comprising modifying an orientation of the AR item based on an orientation of a surface on which the AR item is placed in the real-world environment.

16. The method of claim 1, further comprising: determining a type of the real-world object; determining that the type of the real-world object corresponds to a type of the AR item; and selectively modifying pixels corresponding to the real-world object in response to determining that the type of the real-world object corresponds to the type of the AR item.

17. The method of claim 1, further comprising training a neural network classifier to determine the classification by: receiving training data comprising a plurality of training images and ground truth room classifications for each of the plurality of training images, each of the plurality of training images depicting a different real-world environment classification; applying the neural network classifier to a first training image of the plurality of training images to estimate a real-world classification of the real-world environment depicted in the first training image; computing a deviation between the estimated real-world environment classification and the ground truth real-world environment classification associated with the first training image; updating parameters of the neural network classifier based on the computed deviation; and repeating the applying, computing and updating steps for a set of the plurality of training images.

18. A system comprising: at least one processor configured to perform operations comprising: receiving a video that includes a depiction of a real-world object in a real-world environment; determining, by a trained neural network, a classification for the real-world environment by processing the real-world object depicted in the video; selecting an augmented reality (AR) item based on the classification of the real-world environment and based on the real-world object depicted in the video; modifying pixel data corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object; adding a depiction of the AR item to the modified video at a display position corresponding to the modified pixel data, the adding the depiction of the AR item to the modified video comprising: calculating characteristic points for a set of elements of the real-world object to generate a mesh based on the calculated characteristic points; generating one or more areas on the mesh of the real-world object; aligning a position of the one or more areas of the real-world object with one or more elements of the AR item; and modifying one or more visual properties of the one or more areas to cause a user device to display the AR item within the video at an individual display position relative to the display position of the real-world object that has had pixel data modified; detecting input that moves the AR item to a new position in the video; determining that the AR item in the new position no longer overlaps a portion of the real-world object that has had the corresponding pixel data modified; and in response to determining that the AR item in the new position no longer overlaps the portion of the real-world object that has had the corresponding pixel data modified, undoing modification of the pixel data to return pixel values of the real-world object to an original value in the video.

19. The system of claim 18, wherein the operations further comprise blurring a region corresponding to the modified pixel data, wherein the AR item is added to the blurred region.
20. A non-transitory machine-readable storage medium that includes instructions that, when executed by one or more processors of a machine, cause the machine to perform operations comprising: receiving a video that includes a depiction of a real-world object in a real-world environment; determining, by a trained neural network, a classification for the real-world environment by processing the real-world object depicted in the video; selecting an augmented reality (AR) item based on the classification of the real-world environment and based on the real-world object depicted in the video; modifying pixel data corresponding to the real-world object depicted in the video to generate a modified video that excludes the depiction of the real-world object; adding a depiction of the AR item to the modified video at a display position corresponding to the modified pixel data, the adding the depiction of the AR item to the modified video comprising: calculating characteristic points for a set of elements of the real-world object to generate a mesh based on the calculated characteristic points; generating one or more areas on the mesh of the real-world object; aligning a position of the one or more areas of the real-world object with one or more elements of the AR item; and modifying one or more visual properties of the one or more areas to cause a user device to display the AR item within the video at an individual display position relative to the display position of the real-world object that has had pixel data modified; detecting input that moves the AR item to a new position in the video; determining that the AR item in the new position no longer overlaps a portion of the real-world object that has had the corresponding pixel data modified; and in response to determining that the AR item in the new position no longer overlaps the portion of the real-world object that has had the corresponding pixel data modified, undoing modification of the pixel data to return pixel values of the real-world object to an original value in the video.